Examiner only

5. (a) The first stage in the preparation of a salt from an acid and an alkali often involves carrying out a titration.

A-**F** below are the steps of the method that can be used to prepare a salt from an acid and an alkali. However, these steps are **not** in the correct order.

A Repeat without the indicator using the same volumes of alkali and acid

B Add 2-3 drops of indicator

C Allow the solution to cool and crystallise naturally

D Place the salt solution into an evaporating dish and heat until half the volume of solution remains

E Add the acid to the alkali until the indicator changes colour

F Measure 25.0 cm^3 of alkali into a conical flask

(i) Place the steps of the method into the correct order. The first and last steps have already been added. [2]

F → ☐ → ☐ → ☐ → ☐ → **C**

(ii) Give the **letter** of the step where error is **most** likely to occur. Give the reason for your choice. [2]

Step ....

Reason ....

....

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(b) The reaction between an acid and an alkali produces a salt and water. The following equation shows how potassium sulfate is produced from an acid and an alkali.

potassium hydroxide + ...**?**............ $\longrightarrow$ potassium sulfate + water

(i) Give the name of the missing acid. [1]

....

(ii) Give the chemical formula of potassium sulfate. [1]

....

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